

Project Report

FitQuest AI

An AI-Powered Fitness & Wellness Platform

Move. Compete. Improve.

Field	Detail
Problem Statement ID	SIH26196
Problem Statement Title	Student Innovation: Ideas that can boost fitness activities and assist in keeping fit
Theme	Fitness and Sports
PS Category	Software
Team Name	Flex AI
Team ID	163570
Application	FitQuest AI — web-based, and responsive across mobile and desktop

Executive Summary

FitQuest AI is our entry for Smart India Hackathon 2026, built around the problem statement on student innovation for fitness activities. At its centre is a camera-based AI trainer that tracks exercise in real time, alongside a yoga coach that speaks in three languages, meal and nutrition tracking, a water target tailored to each user, and a full gamification layer — XP, levels, badges, streaks, challenges and a leaderboard. It all lives in one responsive app that works equally well on a phone or a laptop, with no extra hardware needed.

We built it around two things that quietly derail most people's fitness efforts: nobody is around to say whether your form is right, and nothing keeps you coming back once the initial motivation fades. The AI Coach solves the first by running pose-tracking computer vision directly on the user's device, counting reps and scoring form live as they move. The rewards system, friendly competition, and easy progress-sharing tools are our answer to the second — turning an ordinary workout into something closer to a routine people actually want to keep up.

This report covers the working build as it stands today — every screen and feature, how the system fits together technically, why we believe it is feasible and worth building further, how it stacks up against existing fitness apps, and what we're planning next.

1. Problem Statement

Problem Statement ID: SIH26196

Title: Student Innovation: Ideas that can boost fitness activities and assist in keeping fit

Theme: Fitness and Sports | Category: Software

1.1 The Core Problem

Sticking to a fitness routine, and eating reasonably well while doing it, is hard for most people — and usually not for lack of trying. What's actually missing is proper guidance, a way to track progress that means something, and a reason to keep going once the early motivation wears off.

The scale of the problem is well documented: the World Health Organization estimates that 31% of adults and 80% of adolescents worldwide don't meet recommended physical-activity levels, despite the well-established physical and mental health benefits of staying active.

And for anyone who could use a professional's help — a trainer to fix a squat, a nutritionist to sanity-check a diet — the cost puts that out of reach for most students and everyday users.

2. Proposed Solution

2.1 Overview

FitQuest AI brings AI, gamification and real-time tracking together in one place, with the aim of making fitness feel personal, guided and, ideally, a bit more enjoyable than it usually is. Workouts, nutrition, activity tracking, challenges and social competition all sit inside the same app, accessible from any modern browser — phone, tablet or computer.

2.2 How It Works

- AI exercise detection watches the user through the device camera, counts reps automatically, and gives live feedback on form — all processed on-device, so no video is ever sent to a server.

- Workout suggestions are personalised to the user's goal, fitness level and recent activity, and adjust each day based on a short readiness check-in covering sleep, energy and soreness.
- An Exercise Guide walks first-timers through any move with a short two-question onboarding flow, producing a five-step walkthrough, a demo clip, tips suited to their specific concern, and a starter plan — before they even open the camera.
- Yoga AI extends the same real-time tracking to ten standing yoga poses, each named in Hindi, Hinglish and English, so the app works for learners from different linguistic backgrounds.
- AI Nutrition estimates calories and macronutrients from a logged meal or a food photo, and a daily water-intake target is worked out automatically from the user's age and weight.
- XP, levels, badges, streaks, daily challenges and a leaderboard give people a reason to stay consistent, while Share My Progress lets users invite friends and family with an auto-generated progress card.
- A Premium tier adds a goal-based Personal Trainer, an AI Fitness Assistant chat, 7-day AI-generated workout and diet plans, and an uncapped XP Store balance.

2.3 How This Addresses the Problem

- Instead of a single static plan, users get personalised, day-to-day fitness support that adapts as they go — directly targeting the inconsistency problem.
- Real-time AI exercise detection and spoken feedback fill the guidance gap, correcting form as it happens rather than after the fact.
- Pairing AI with gamification makes the experience more personal and engaging, which is exactly where most ad-hoc fitness apps lose people.
- Personalised workouts and nutrition tracking point users toward their specific goal, whether that's losing weight, gaining weight, or building muscle.
- XP, challenges and achievements give people a reason to open the app again tomorrow — the consistency that most fitness apps struggle to sustain.
- Step tracking, Yoga AI and wearable integration extend the platform past structured workouts into everyday activity.

3. Complete Feature Set

The rest of this section documents the current working build module by module, expanding on the summary above with the full functional detail of what's actually implemented.

3.1 Onboarding & Authentication

- Sign-up captures the details we use to personalise the app: age, weight, height, gender, fitness goal and activity level.
- Standard login, sign-up and forgot-password flows, plus a one-tap "Continue as Demo Athlete" mode so anyone can try the app without creating an account.
- Every personalisation figure — daily water target, calorie and macro targets, suggested workout intensity — recalculates automatically whenever the user updates their profile.

3.2 Dashboard

The dashboard is the user's daily home screen, and it adapts to their profile and recent activity each time it loads.

- Daily Water Target — shown right at the top, calculated from the age and weight given at sign-up (roughly 40 ml/kg under age 30, 35 ml/kg between 30 and 55, and 30 ml/kg over 55, rounded and capped to a sensible range).
- Morning Readiness Check-in — three quick taps covering sleep, energy and soreness, which adjust the day's suggested workout intensity.
- Today's Suggested Workout — a short routine generated from the user's goal, fitness level and readiness score.
- Weekly Streak and Weekly Workout Split — a quick visual read on consistency and how workout types have balanced out over the past week.
- Step Tracker — automatic step and distance tracking using the phone's motion sensor, no extra app needed.
- Daily Spin Wheel — a once-a-day spin for a bit of bonus XP or XP Coins.
- Current Challenge and Leaderboard Position — where the user stands right now.
- Progress Analytics preview and Recent Achievements — the latest badges and milestones.
- Connected Device card — pairs a Bluetooth heart-rate strap or smartwatch, so live heart rate feeds into AI Coach sessions and calories are calculated from real heart-rate data.

3.3 AI Coach

AI Coach is the heart of the platform — real-time exercise tracking across 12 strength and cardio exercises:

Exercise	Level	Category	Muscles Worked
Squat	Beginner	Lower Body, Strength	Quads, Glutes, Hamstrings
Push-up	Intermediate	Upper Body, Strength	Chest, Triceps, Shoulders
Jumping Jack	Beginner	Cardio, Full Body	Full Body
Lunge	Intermediate	Lower Body, Strength	Quads, Glutes
Plank (hold)	Beginner	Core, Full Body	Core, Shoulders
Sit-up	Beginner	Core	Abs
High Knees	Beginner	Cardio, Lower Body	Hip Flexors, Core
Bicep Curl	Beginner	Upper Body, Strength	Biceps
Shoulder Press	Intermediate	Upper Body, Strength	Shoulders, Triceps
Glute Bridge	Beginner	Lower Body, Strength	Glutes, Hamstrings
Calf Raise	Beginner	Lower Body	Calves
Mountain Climber	Advanced	Cardio, Core, Full Body	Core, Shoulders, Hip Flexors

- Live, camera-based pose tracking runs entirely on-device using 33 body landmarks, turning joint positions into joint angles for each exercise's specific check.
- Automatic rep counting and a live 0–100 form score, with corrective feedback spoken aloud the moment a fault is caught — knees caving in on a squat, hips sagging in a plank, that kind of thing.
- Hold-based exercises like the plank track hold time on a timer instead of counting reps.

- Session stats — reps, form score, calories, duration — are logged for that workout and feed into the dashboard, XP, streak and progress-analytics systems.
- A connected wearable supplies live heart rate during a session, sharpening the calorie estimate for that workout.

3.4 Exercise Guide — Guided Onboarding for Beginners

Any exercise or yoga pose can be opened as a short, self-contained guide for someone who's never tried it, without needing to start the camera first.

- Two quick questions — prior experience with the exercise, and the user's specific worry (balance, knee discomfort, wrist strain, that sort of thing) — shape the guide that follows.
- A five-step walkthrough with a looping demo clip (or, where no clip exists yet, an icon panel showing muscles worked and the key form point), a Back/Next stepper, and optional Read Aloud narration.
- Tips matched to whatever the user is worried about, plus a starter plan — sets, reps or hold time — suited to their experience level.
- A "Practice with AI camera" button hands off straight into AI Coach for that same exercise.

3.5 Yoga AI

Yoga AI applies the same real-time pose-tracking engine to ten standing yoga poses, each named in three languages so the app works for Hindi, Hinglish and English speakers alike:

Hindi Name	Hinglish Name	English Name	Level
वृक्षासन	Vrikshasana	Tree Pose	Beginner
वीरभद्रासन 2	Veerbhadrasana 2	Warrior II	Intermediate
उत्कटासन	Utkatasana	Chair Pose	Intermediate
ताड़ासन	Tadasana	Mountain Pose	Beginner
ऊर्ध्व हस्तासन	Urdhva Hastasana	Upward Salute	Beginner
वीरभद्रासन 1	Veerbhadrasana 1	Warrior I	Intermediate
उत्कट कोणासन	Utkat Konasana	Goddess Pose	Intermediate
त्रिकोणासन	Trikonasana	Triangle Pose	Intermediate
ताड़ासन	Tiryak Tadasana	Standing Side Bend	Beginner
प्रणामासन	Pranamasana	Prayer Pose	Beginner

- Each pose has its own alignment check — stance width and knee angle for Goddess Pose, torso tilt and arm position for Triangle Pose, and so on — so hold time only counts while the pose is actually correct.
- The same Exercise Guide, demo panel and AI Coach hand-off available for regular exercises are available for every yoga pose too.

3.6 Exercise Library

- A searchable, filterable catalogue of all 22 exercises and poses, searchable by English, Hindi or Hinglish name.
- Each entry shows difficulty, category, muscles worked, and one-tap access to both the AI Coach session and the Exercise Guide.

3.7 Nutrition AI

- Daily calorie and macronutrient (protein / carbs / fat) targets, calculated from BMR and TDEE using the Mifflin-St Jeor equation, based on weight, height, age and gender.
- A meal log split across Breakfast, Lunch, Snack and Dinner, tracked against those daily targets.
- A food-photo upload option that returns an estimated food name, calories and macronutrients — recognition is simulated in the current prototype build.
- The Daily Water Target shown on the dashboard is calculated and pulled from this same profile data.

3.8 Personal Trainer & AI Fitness Assistant (Premium)

- Personal Trainer — goal-based programs for weight loss, weight gain and muscle building, generated from the user's profile and updated as their weight and goal change.
- A 7-day AI-generated workout and diet plan, refreshed on request.
- AI Fitness Assistant — a built-in chat that answers common fitness and nutrition questions instantly, on things like muscle soreness or building muscle, without needing a human coach.

3.9 Gamification: XP, Levels, Badges, Streaks & Challenges

- Every workout, yoga session and logged activity earns XP, which levels the user up through five tiers: Beginner, Active, Fit, Athlete and Fitness Master.
- Seven badges reward specific milestones — First Workout, 7-Day Streak, 100 Rep Warrior, Speed Champion, Perfect Form, Challenge Champion and Fitness Master.
- A daily streak counter and a weekly activity split give a quick visual read on consistency.
- A Daily Spin Wheel grants a once-a-day bonus of XP or XP Coins.
- Challenges are time-boxed goals — a rep count or step count target, say — that track progress, show how many people are taking part, and pay out an XP reward on completion.
- A Leaderboard ranks users against the wider community for a bit of social motivation.

3.10 XP Store

- XP Coins are earned through everyday app use and can be redeemed in the XP Store for vouchers, and for Premium members, real fitness products.
- The conversion rate is fixed at 50 XP Coins to ₹1, so every action in the app has a transparent, real-world value.
- Free accounts keep a spendable balance capped at 3,000 XP Coins — earning XP and levelling up are never capped, only the redeemable store balance — while Premium accounts have no cap at all.

3.11 Share My Progress

- Generates an on-device progress card showing the user's level, XP, streak, workout count, total reps, calories burned, the week's activity, and badges earned.

- Produces an editable invite message and shares the card directly via WhatsApp, the device share sheet, or copy-to-clipboard — or the user can just download the image.
- Built specifically to help users bring friends and family into the app, and rewards a daily XP bonus for sharing (shared with the existing Share & Earn bonus, so it can't be claimed twice in one day).
- The card and message only ever include fitness statistics — never weight, age, email or other private profile details.

3.12 Progress Analytics, Reminders & Data

- Progress Analytics charts reps, XP, calories and form score over time.
- Workout Reminders can be scheduled to prompt regular sessions.
- Data Storage — all progress, logs and settings are stored locally on the user's device (browser local storage), with an export option, and camera video for pose tracking is processed on-device and never permanently stored.

3.13 Premium Plans & Pricing

FitQuest AI runs on a freemium model. Everyone gets the core AI Coach, Yoga AI, challenges and community access for free; Premium unlocks the fuller coaching and planning suite.

Plan	Price	What's Included
Free	₹0 / forever	Basic workouts & challenges, community access, limited analytics
Premium	₹299/month (₹179/month billed yearly — save 40%)	Personalised 7-day AI workout & diet plan; AI Fitness Assistant chat; Personal Trainer programs; calorie & macro targets from BMR/TDEE; weight progress tracking; Bluetooth watch & heart-rate integration; uncapped XP Coin balance

4. Technical Approach

4.1 Technology Stack

Layer	Details
Frontend	HTML, CSS3, JavaScript — a responsive UI built for both PC and mobile
Backend	FastAPI, NVIDIA API, Resend API
Database	SQLite (local machine)
Hosting	Ngrok public URL (current prototype)
AI / Computer Vision	MediaPipe Pose — real-time camera input, 33 body landmarks, joint-angle calculation, exercise-specific detection, rep counting, form scoring
Browser / Device APIs	Camera API, Device Motion API, Web Bluetooth API, Speech Synthesis API, Web Share API
Data	JavaScript state management, browser LocalStorage, JSON data storage / export

4.2 Architecture

The app is organised into four layers that together form the AI Coach pipeline:

- User Interface — login, sign-up, forgot password, authentication and access, onboarding and profile setup.
- Dashboard / Monitoring & Results — workouts and schedule, Nutrition AI, challenges, leaderboard and progress analytics.
- AI Fitness Processing — camera and live video feed into on-device pose detection, then exercise analysis, rep counting and form scoring, and finally real-time feedback.
- Personalization & Gamification — XP, badges and streaks, backed by local data covering browser camera access, local state and logs, nutrition data, fitness-logic rules, and challenge/XP state.

The AI Coach pipeline itself runs end-to-end on the user's device: select an exercise, grant camera permission, start the live camera feed, run pose detection, analyse the exercise, count reps and score form, give real-time feedback, and update XP and progress once the workout is complete.

On privacy: camera video is processed locally on the user's device and is never permanently stored.

5. Feasibility and Viability

5.1 Technical Feasibility

- Runs entirely as a web-based application
- Camera-based pose detection
- Computer vision for exercise analysis
- Works with a standard smartphone or PC camera
- Device motion sensing for step tracking
- Wearable connectivity
- Local, user-controlled data management
- Real-time progress tracking

5.2 Operational Feasibility

- Minimal equipment required
- Quick onboarding and goal setup
- Straightforward, easy-to-use interface
- A genuinely gamified experience
- Works for both individual and group participation
- Suited to home or gym use alike

5.3 Limitations and How We're Addressing Them

Limitation	Mitigation Strategy
Camera angle & lighting	On-screen guidance for positioning the camera correctly
Pose visibility / occlusion	Visibility and confidence checks before tracking begins
Food recognition accuracy	Manual correction / user confirmation
Device compatibility	Verified nutrition data and device compatibility checks

Internet dependency for web services	Ongoing model and feature improvements
AI recommendations need validation	Manual correction / user confirmation

5.4 Viability

- Low initial infrastructure cost
- Modular, expandable architecture
- A large potential user base
- Freemium / Premium revenue model
- XP Store and fitness rewards drive engagement
- Easy to expand with new features
- A scalable web platform

6. Impact and Benefits

Improved Health & Fitness — guided fitness becomes accessible to more people, at whatever level they're starting from, with better outcomes as a result.

Better Nutrition & Lifestyle — tracking calorie and nutrient intake nudges users toward healthier eating habits over time.

Mental Well-being & Motivation — challenges, streaks and achievements give people a reason to stay engaged, which matters for motivation as much as fitness itself.

Community & Social Connection — users can compete and collaborate with others, and bring friends and family in through Share My Progress.

Encourages Consistency — progress tracking and achievements push toward regular fitness habits rather than occasional, one-off sessions.

Personalized Workouts — suggestions are shaped by the user's goals, fitness level and how ready they feel that day.

AI Exercise Detection — exercises are recognised in real time, with reps counted automatically and form corrected as it happens.

Nutrition Awareness — logged meals and food photos are analysed to estimate calories, protein, carbs and fat.

AI Coach — accessible fitness guidance without needing to hire a human personal trainer.

Gamification — XP, coins, challenges and rewards keep daily fitness feeling less like a chore.

7. Research and References

7.1 Research

- The WHO reports that 31% of adults and 80% of adolescents worldwide don't meet recommended physical-activity levels.
- The WHO also notes that regular physical activity brings significant physical and mental health benefits.
- WHO guidance on healthy diets emphasises adequacy, balance, moderation and diversity.

- Google's ML Kit / MediaPipe documentation describes pose detection using 33 body landmarks and confidence scores — this is what underpins both AI Coach and Yoga AI.

7.2 References

WHO — Physical Activity: <https://www.who.int/news-room/fact-sheets/detail/physical-activity>

WHO — Healthy Diet: <https://www.who.int/news-room/fact-sheets/detail/healthy-diet>

Google ML Kit / MediaPipe Pose — Pose Detection: <https://developers.google.com/ml-kit/vision/pose-detection>

Google — Pose Detection for Android: <https://developers.google.com/ml-kit/vision/pose-detection/android>

7.3 How FitQuest AI Compares

Feature	FitQuest AI	Nike Training Club	Fitbod	MyFitnessPal
AI Exercise Detection & Rep Counting	Yes	No	No	No
AI Food Image / Nutrition Analysis	Yes	No	No	Yes
Personalized Workout Plans	Yes	Yes	Yes	No
AI Coach / Guided Fitness	Yes	Yes	Yes	Yes
Progress & Activity Analytics	Yes	Yes	Yes	Yes
Challenges & Daily Goals	Yes	Yes	Yes	No
Leaderboard / Group Competition	Yes	No	No	No
XP / Coins / Rewards	Yes	Yes	No	No
Personal Trainer	Yes	No	Yes	No
Premium Fitness Ecosystem	Yes	Yes	Yes	Yes

8. Future Scope

The current build already covers the core AI coaching, yoga, nutrition and gamification experience end to end. Here's what we're planning for upcoming releases.

8.1 Zumba AI

- A dedicated Zumba AI mode is planned, using the same on-device pose-tracking engine to follow dance-style cardio routines.
- Rather than scoring joint angles against a fixed pose, it will measure rhythm and movement consistency — how evenly spaced a user's repeated moves are — to give an "on-beat" accuracy score suited to free-form dance.
- It's planned to sit alongside AI Coach and Yoga AI as a third guided-tracking mode, with its own exercise library entries, XP rewards and Exercise Guide onboarding.

8.2 Report Analysis for Suggested Exercises

- A planned analytics engine that looks at a user's historical session data — form scores, completed exercises, streak patterns and readiness check-ins — to suggest which exercises or routines to focus on next.
- For example, someone with a consistently lower form score on lower-body exercises would get more squat, lunge or glute-bridge sessions suggested, along with easier variants or the Exercise Guide, until their form improves.
- The same analysis should eventually personalise the dashboard's suggested workout beyond the current goal-and-readiness model, closing the loop between tracked performance and future recommendations.

8.3 Expanded Premium Options

- Additional Premium tiers and add-ons are planned, to give users more granular choices depending on their needs and budget.
- This includes a wider real-product catalogue in the XP Store, deeper analytics and export options, early access to new AI features like Zumba AI and Report Analysis, and family or group Premium plans that let linked accounts share benefits.

8.4 Other Roadmap Items

- Moving from local SQLite storage to a cloud-hosted backend and a proper domain, enabling multi-device sync beyond the current Ngrok-based prototype.
- Replacing simulated food-photo recognition with a production computer-vision model.
- Native iOS and Android apps alongside the current responsive web app.
- Extending trilingual naming beyond Yoga AI to the full exercise library, and adding further regional languages.

9. Conclusion

FitQuest AI turns a phone or laptop camera into a real-time personal trainer, pairs it with an approachable, trilingual yoga program and nutrition tracking, and wraps the whole thing in a gamified, social layer that gives people an actual reason to come back the next day. The result is a single, accessible platform that speaks directly to problem statement SIH26196 — helping students and everyday users build, and keep, a consistent, guided fitness habit, without the cost of a personal trainer.

With Zumba AI, performance-based exercise recommendations, and an expanded Premium tier already on the roadmap, we've built FitQuest AI as a modular platform with plenty of room to grow well beyond this first working build.